

Sales Prediction Management System

Forecasts Page — Full Study Guide

[Generate forecast](#) · [Chart](#) · [Inventory suggestion](#) · [Reorder watchlist](#) · [MAE / RMSE](#)

Plain-English notes with examples for self-study and thesis defense preparation.

How to use this PDF

Part A = whole Forecasts page tour. Part B = deep dive on inventory numbers (stock, predicted demand, reorder level, suggest order) and MAE/RMSE. Read Part B slowly with a calculator if you want. End has quiz + answers.

PART A — What is the Forecasts page?

The Forecasts page is the **manager planning board**. The shop already records sales and stock. Here you ask: *How much of this product might we sell soon, and do we have enough stock?*

Only **Manager** and **Admin** can open it. Staff use New Sale / History instead.

You (manager) click Run prediction → PHP calls Python brain → brain returns next N days of units → saved in MySQL → chart + tables + reorder tips.

A1. Online / offline banner

Banner	Meaning
Prediction engine online	Python Flask brain is running. You can run new predictions.
Prediction service offline	Brain not running. Old saved forecasts may still show; new runs fail until you start python app.py.

A2. Generate forecast (left form)

Control	Meaning
Product	Which item to forecast (Tissue Pack, Indomie, milk power, ...). Units always mean units of THIS product.
Model	Who guesses: LSTM (sequence model) or Linear Regression (simple baseline).
Horizon (days)	How far ahead. Horizon 7 = next 7 days.
Run prediction	Ask the brain now; save results; refresh chart/tables.
Update model (retrain)	Admin only. Teach the brain again from newest sales. Then run prediction again.

A3. History vs forecast (chart)

Line	Meaning
Blue solid — Actual sales	What really sold in the past (~last 45 days). Truth.
Orange dashed — Predicted	What the model guesses for future days. Not real sales yet.

Left side of the chart = past (blue). Right side with future dates = guess (orange). Small text above may show model name, MAE, RMSE, and when the forecast was generated.

A4. Saved forecast rows (table)

Same forecast as the orange line, listed day by day.

Column	Meaning
Date	Future calendar day being guessed
Predicted qty	How many units of the SELECTED product expected that day
Model	Istm or linear_regression
MAE / RMSE	Model quality scores from evaluation (lower is better)

A5. Reorder watchlist

Shop-wide view: for each product, compare latest saved forecast total vs current stock. Red **Short** = may need restock. Green **OK** = stock covers predicted demand. If you never ran a forecast for a product, predicted demand may be 0.

A6. Recent model training scores

Report card from training (python train.py or Update model). Compare linear_regression_p3 vs Istm_p3: lower MAE/RMSE usually means better on the test period.

PART B — Inventory suggestion explained carefully

This is the section people confuse most. Read slowly.

B1. Everything is about ONE product

When you select **Tissue Pack** and run prediction, every number on that panel is about **Tissue Pack units only** — not money, not all products mixed.

- bullet Predicted qty = how many **packs of tissue** (if product is Tissue Pack)
- bullet If product is Indomie Carton → units = cartons of Indomie
- bullet If product is milk power → units = packs/tins of that product
- bullet Stock = how many of that same product you have in the store now

Remember

Predicted quantity is always: units of the product you selected — not naira, not mixed items.

B2. The four numbers (definitions)

Term	What it is	What it is NOT
Current stock	How many units you have on the shelf / in store RIGHT NOW	Not future stock. Not sales money.
Predicted demand (horizon)	SUM of predicted daily sales over the next N days (e.g. 7 days). How many units customers may BUY in that period.	NOT 'stock you will have in 7 days'. NOT leftover inventory.
Static reorder level	A fixed warning line YOU set on the product (e.g. 10). When stock ≤ 10 , dashboard can show low-stock alert. Manual rule.	NOT the AI forecast. NOT automatic order quantity.
Suggest order	Extra units to BUY from supplier so stock can cover predicted demand. Roughly: demand – stock (if demand > stock).	NOT 'you will have 35 in stock'. It is 'you may need to order 15 more'.

B3. Your example — fixed step by step

You said: stock = 20, next 7 days predicted total = 35, extra 15, suggest order = ?

Correct reading

- bullet Stock **now** = 20 units of that product (e.g. 20 tissue packs).
- bullet Horizon = 7 days.
- bullet Predicted demand = 35 means: over the next 7 days, the model thinks customers will buy about **35 units total** (day1+day2+...+day7).
- bullet That is **expected sales / demand**, not 'stock level after 7 days'.
- bullet If nobody restocks: 20 on hand is less than 35 needed → you may run short by about 15.
- bullet Suggest order $\approx 35 - 20 =$ **15 units to buy from supplier**.

Wrong reading (common mistake)

Wrong: "In 7 days my stock will become 35." Right: "Customers may want about 35 units in 7 days; I only have 20; I should order about 15."

B4. Mini table for the same example

Day	Predicted qty (units of THIS product)	Running total demand
Day 1 (e.g. Mon)	6	6
Day 2	5	11
Day 3	4	15
Day 4	5	20
Day 5	6	26

Day	Predicted qty (units of THIS product)	Running total demand
Day 6	4	30
Day 7	5	35

Predicted demand (horizon) = 35. Current stock = 20. Suggest order = $\text{ceil}(35 - 20) = 15$.

If after 7 days you sold exactly 35 and never restocked, stock would go toward zero (or negative in theory — the system blocks overselling on real sales). The suggestion is meant to help you restock **before** that happens.

B5. Another example (enough stock)

- Product: Indomie Carton
- Current stock = 100 cartons
- Horizon 7 predicted demand = 40 cartons
- $100 \geq 40 \rightarrow$ green: stock looks enough
- Surplus $\approx 100 - 40 = 60$ cartons buffer
- Suggest order = 0 (no forced order from this rule)

B6. Static reorder level vs predicted demand

These are two different alarms.

Alarm	Based on	Example
Static reorder level	Fixed number on product form (manager sets 10, 20, ...)	Stock is 8 and reorder level is 10 $\rightarrow$ low-stock badge on dashboard even without AI
Predicted demand / suggest order	AI forecast for next N days vs current stock	Stock 50, reorder level 10 (looks fine statically), but 7-day demand is 80 $\rightarrow$ still suggest order 30

So: reorder level = simple permanent warning line. Predicted demand = smart short-term expected sales. A good manager looks at both.

B7. Where the daily predicted qty comes from

- Model reads recent history of that product (and features like weekend/promo).
- Model outputs a number for each future day: e.g. 6, 5, 4, ...
- Those appear in Saved forecast rows and as the orange chart line.
- Inventory suggestion adds those days together for the horizon total.

Formula used in the app (conceptually):

```
predicted_demand = sum(predicted_qty for each day in horizon) if predicted_demand > stock: suggested_order = ceil(predicted_demand - stock) else: suggested_order = 0
```

PART C — MAE and RMSE explained better

MAE and RMSE are **report card scores** for the model. They are NOT sales forecasts and NOT stock levels.

One line to memorize

MAE / RMSE measure how wrong the model was on past test days. Lower number = better. They do not tell you how many units to order.

C1. Why we need them

Anyone can invent a guess. Metrics ask: when we hide the last part of history and let the model predict those days, how close was it to the truth?

C2. Absolute error (building block)

For one day: absolute error = |true sales – predicted sales| (always positive distance).

True = 10 tissue packs, predicted = 7 → error = 3 packs that day.

C3. MAE — Mean Absolute Error

MAE = average of those absolute errors across many test days.

Manager English: “On a typical test day, the model was off by about MAE units.”

Example days true: 10, 8, 12. Predictions: 9, 10, 11. Errors: 1, 2, 1. MAE = $(1+2+1)/3 = 1.33$ units.

If product is Tissue Pack and MAE = 1.33, the model was wrong by about 1.3 packs per day on average on the test set — not 1.3 naira.

C4. RMSE — Root Mean Square Error

RMSE also measures average error, but **big mistakes hurt more** because errors are squared first.

- Square each error: $1^2=1$, $2^2=4$, $1^2=1$
- Mean of squares = $(1+4+1)/3 = 2$
- RMSE = $\sqrt{2} \approx 1.41$ units

If most days are slightly wrong but one day is wildly wrong, RMSE rises more than MAE. That helps you see unstable models.

C5. MAE vs RMSE cheatsheet

	MAE	RMSE
Idea	Average absolute miss	Average miss, big misses amplified
Unit	Same as product units	Same as product units
Lower better?	Yes	Yes
Sensitive to outliers?	Less	More
Manager phrase	Usually off by ~MAE units/day	Penalizes large bad days more

C6. Worked comparison (Linear vs LSTM)

Model	MAE	RMSE	How to read
Linear Regression	2.27	2.91	On average ~2.3 units wrong/day on test
LSTM	1.86	2.33	On average ~1.9 units wrong/day — better here

Because LSTM MAE and RMSE are lower, we say LSTM performed better on that product's test period. That does not guarantee next week will be perfect — only that it fit history better.

C7. What MAE/RMSE are NOT

- bullet NOT the number of units to order
- bullet NOT predicted demand for 7 days
- bullet NOT stock remaining
- bullet NOT a percentage unless you convert them yourself
- bullet NOT comparable blindly across products with very different sales scales (a MAE of 2 on USB sticks vs 2 on rice bags is different business impact)

PART D — Full walkthrough examples

Example 1 — Tissue Pack

- You select product: Tissue Pack
- Horizon: 7, Model: LSTM
- Run prediction
- Saved rows might show Mon 3, Tue 2, Wed 4, Thu 3, Fri 5, Sat 6, Sun 5 → demand = 28 packs
- Current stock = 12 packs
- $28 > 12$ → Suggest order = 16 packs of tissue
- Static reorder level = 20 → even without AI, stock 12 is already below 20 (low stock alert)
- MAE 1.88 means historically the model was off by ~2 packs/day on average on test days

Example 2 — Rice 50kg (slow mover)

- Predicted 7-day demand = 8 bags total
- Stock = 40 bags
- $40 > 8$ → OK, suggest order 0
- Reorder level = 15 → stock still above static warning
- Manager may still order early for supplier lead time — AI is advice, not a law

Example 3 — Promo week

- Product has an active promotion in the Promotions table
- Feature is _promo helps the model expect higher sales
- Predicted demand may jump vs a normal week
- Inventory suggestion may flip from OK to Short even if stock looked fine last week

PART E — How sections connect

Generate forecast → creates daily predicted qty Chart → draws history (blue) + predictions (orange) Saved rows → same predictions as a table Inventory suggestion → $\text{sum}(\text{predictions})$ vs stock for THIS product Reorder watchlist → same idea for many products Training scores → MAE/RMSE quality of brains when trained

PART F — Self-check quiz

1. If predicted demand (horizon 7) is 35, does that mean stock will be 35 in 7 days?
2. Stock 20, demand 35. What is suggest order about?
3. Predicted qty is units of what?
4. What is static reorder level?
5. MAE = 2 for Tissue Pack means what in plain English?
6. Why might RMSE be higher than MAE?
7. Blue chart line vs orange chart line?
8. Can Staff open Forecasts?

Answers

1. No. 35 is expected customer purchases over 7 days, not ending stock.
2. About 15 more units to buy so stock can cover demand (35–20).
3. Units of the selected product only (e.g. tissue packs if Tissue Pack is selected).
4. A fixed low-stock warning number set on the product; not the AI demand total.
5. On test days the model was wrong by about 2 packs per day on average.
6. Because large mistakes are squared and pull RMSE up more.
7. Blue = real past sales; orange = model's future guess.

8. No — Manager/Admin only.

PART G — Study tips

- Explain Example B3 out loud without looking (stock 20, demand 35, order 15).
- On the live page, pick one product and write: stock, demand, suggest order, MAE.
- Never mix MAE with suggest order — different jobs.
- Defense sentence: “We forecast unit demand over a horizon and compare it to on-hand stock to support reorder decisions; model quality is reported via MAE and RMSE.”

Commands (if services stopped)

Shop: `http://localhost/spms/` (Apache + MySQL in XAMPP)

Brain: `cd ml_service → python app.py` or `docs/start-ml.ps1`

Login `manager@spms.local / password123 → Forecasts`

End of guide. Re-read Part B and Part C before demos. Your ML theory PDF (Linear Regression & LSTM) pairs well with this UI guide.